iting further enhancement. Conversely, traits conflicting with safety training (e.g., *self-interested*) exhibit strong resistance to activation even at high coefficients, reflecting the model's alignment constraints. To ensure evaluator objectivity, we validate these findings using multiple independent LLM judges from different organizations in Appendix A.5. Beyond correlational analysis, we establish causal independence through controlled multi-trait interventions with cross-layer verification in Appendix A.6. To address potential evaluation circularity concerns, we validate our LLM-based judges against human expert ratings and external judges in Appendix A.7. We quantify these alignment-induced effects through activation success metrics and evaluate their impact on model safety using adversarial benchmarks in Appendix A.9

Beyond effectiveness validation, we assess the orthogonality of persona vectors to verify they represent independent personality dimensions. Figure 2 reveals that opposing trait pairs exhibit strong negative cosine similarities, confirming their antithetical nature. Interestingly, some cross-dimensional correlations reflect semantic associations in the model's training data. For instance, the positive similarity between *nervous* and *careless* likely stems from textual patterns where anxiety co-occurs with descriptions of reduced attention or motor control, demonstrating how the model's learned representations capture psychological associations present in natural language. We present more analysis on these unexpected correlations in Appendix A.10. Importantly, we demonstrate that these non-orthogonal correlations do not compromise algebraic operations; secondary effects remain predictable and compositional, as validated in Appendix A.11.

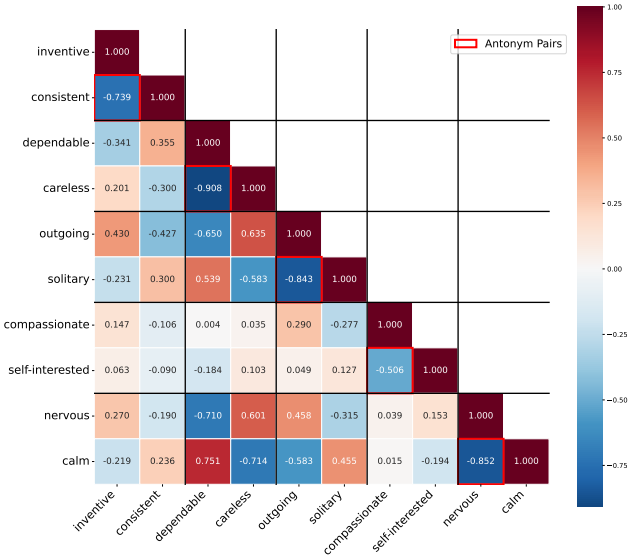

Figure 2: Cosine similarity between persona vectors.

2.3 PERSONA-ALGEBRA: COMPOSITIONAL OPERATIONS

PERSONA-ALGEBRA demonstrates that persona vectors form a coherent algebraic system supporting standard mathematical operations. The key insight is that if persona vectors truly encode personality as compositional features, then vector arithmetic should produce predictable changes in personality expression.

2.3.1 EVALUATION METHODOLOGY

To validate this hypothesis, we employ the BFI-44 questionnaire as our evaluation framework. Our validation approach follows three steps: (1) apply vector operations (scalar multiplication, addition, subtraction) to steer the model, (2) have the steered model complete the BFI-44 questionnaire, and (3) verify whether the resulting personality scores align with the expected outcomes of the vector operations. For instance, adding $v_{outgoing} + v_{compassionate}$ should yield high scores in both Extraversion and Agreeableness dimensions.

Since the original BFI-44 contains subjective self-report items unsuitable for LLMs, we adapt it for behavioral evaluation. Each question is transformed into scenario-based prompts using GPT-4o (Details in Appendix A.12), and model responses are evaluated by GPT-4.1-mini on a 5-point Likert scale (Details in Appendix A.12). This behavioral assessment is crucial as LLMs' self-reported traits often diverge from their actual behavioral patterns (Han et al., 2025).

2.3.2 VALIDATION RESULTS

Following our three-step validation approach, we test whether vector operations produce predictable BFI-44 scores. For each operation, we steer the model, administer the adapted BFI-44 questionnaire, and compare the resulting personality scores against expected outcomes.

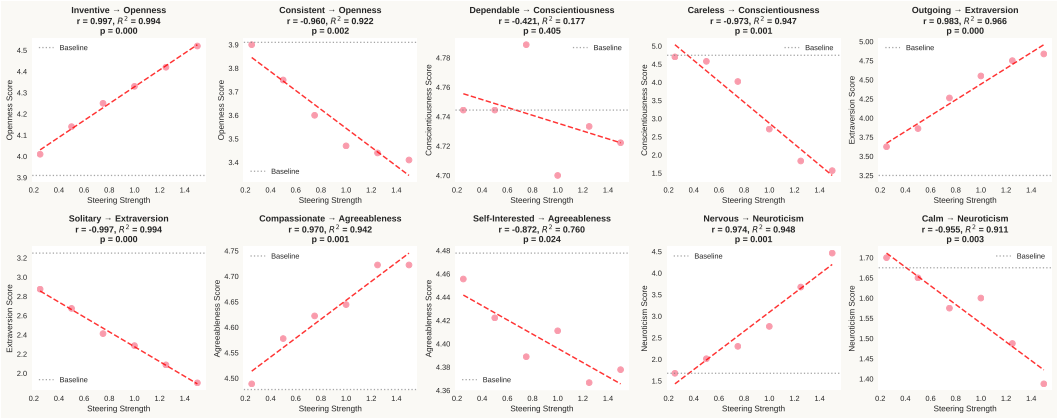

Figure 3: Linear relationship between steering coefficients and BFI dimension scores. All vectors except *dependable* show strong linear modulation (high R²). We conjecture the *dependable* vector's saturation stems from baseline model optimization for conscientiousness.

Scalar Multiplication. We first examine whether multiplying persona vectors by scalar coefficients produces proportional changes in BFI-44 scores. Figure 3 shows the relationship between steering coefficients (α from -1 to 2) and corresponding dimension scores. The results confirm strong linearity: increasing α for $v_{outgoing}$ proportionally increases Extraversion scores, with Pearson correlations exceeding 0.9 for most traits. The *dependable* vector shows saturation in Conscientiousness scores, which we hypothesize reflects the model's baseline optimization for reliability.

Vector Addition and Subtraction. We next test whether vector arithmetic produces the expected combinations of personality traits in BFI-44 scores. Figure 4 presents the results. For vector addition, steering with $v_{outgoing} + v_{compassionate}$ yields high scores in both Extraversion and Agreeableness dimensions, confirming trait composition. For vector subtraction, $v_{outgoing} - v_{solitary}$ amplifies Extraversion scores beyond using $v_{outgoing}$ alone, while $v_{outgoing} - v_{compassionate}$ maintains high Extraversion but reduces Agreeableness, demonstrating trait isolation. These BFI-44 score patterns validate that persona vectors form a coherent algebraic system where standard vector operations translate to predictable personality modifications.

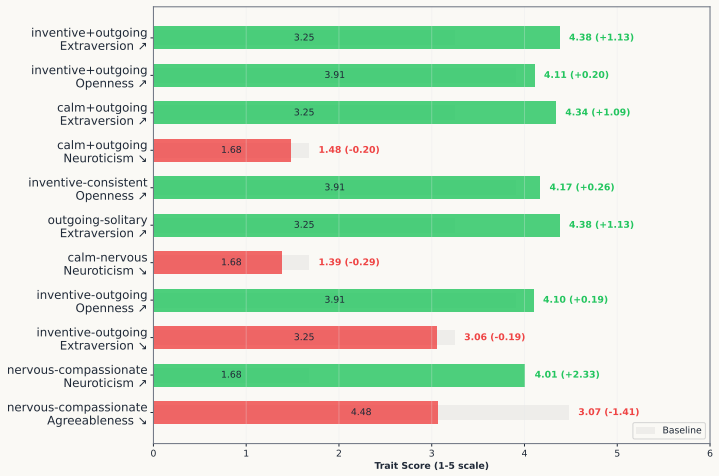

Figure 4: BFI-44 score changes after vector arithmetic operations. Y-axis shows operation and target dimension with expected direction (arrows). Grey: baseline scores; colored: post-steering scores (green for expected increases, red for decreases).

2.4 PERSONA-FLOW: DYNAMIC PERSONALITY CONTROL

PERSONA-FLOW extends the static vector operations validated in PERSONA-ALGEBRA to enable dynamic personality adaptation during inference. While previous components demonstrate that persona vectors support algebraic composition, PERSONA-FLOW applies these operations adaptively based on conversational context. This enables real-time personality modulation without predefined scripts, computing optimal personality configurations through contextual analysis.

2.4.1 MOTIVATION

The predict-then-steer mechanism enables context-aware personality modulation by combining the algebraic properties of persona vectors with dynamic coefficient prediction. This approach allows fine-grained control: enhancing traits relevant to the situation while suppressing conflicting ones. The system can be implemented through various architectures including tool learning (Qin et al., 2024) or direct integration into the generation pipeline, providing flexibility for different deployment scenarios while maintaining the core benefit of adaptive personality control.

2.4.2 METHODOLOGY

We implement dynamic personality control through a predict-then-steer mechanism operating on each conversational turn. This two-stage approach first determines necessary personality adjustments, then applies them through vector steering.

Stage 1: Contextual Personality Prediction. Before generating each response, the model analyzes the conversational context to determine appropriate personality adjustments. Given the persona description and user input, an intermediate inference pass predicts steering coefficients for each dimension. These coefficients specify incremental adjustments (from -2 to +2) that align personality expression with situational demands. Details of the prompt are provided in Appendix A.13. The computational overhead of this predict-then-steer mechanism is analyzed in Appendix A.17.

Stage 2: Vector Composition and Steering. The predicted coefficients are applied to the corresponding persona vectors from PERSONA-BASE. For coefficients exceeding a threshold ($|\alpha| > 0.5$), we compute a composite steering vector: $v_{composite} = \sum_{i \in OCEAN} \alpha_i \cdot v_i$, where α_i represents the predicted coefficient for dimension i. This composite vector is then injected into the model's residual stream at the optimal layer during response generation, following the activation steering methodology established in PERSONA-ALGEBRA. Pseudo-code is provided in Appendix A.16

2.5 PERSONA-EVOLVE: EVALUATION FRAMEWORK

PERSONA-EVOLVE provides a systematic benchmark for evaluating dynamic personality adaptation in conversational contexts. While existing benchmarks assess static personality traits, PERSONA-EVOLVE evaluates whether models can maintain coherent personas while adapting to evolving emotional states and situational demands.

2.5.1 MOTIVATION

Current personality evaluation benchmarks primarily focus on static trait assessment through questionnaires or single-turn interactions. However, real-world conversations require dynamic personality adaptation: maintaining consistent core traits while adjusting emotional expression to situational contexts. PERSONA-EVOLVE addresses this gap by providing multi-turn dialogue scenarios where models must balance persona consistency with contextual appropriateness.

2.5.2 BENCHMARK CONSTRUCTION

We construct PERSONA-EVOLVE through a systematic five-stage pipeline. First, we generate diverse personas representing various professional and personal roles (e.g., Empathetic Family Doctor, Food Truck Owner) that define stable behavioral constraints persisting throughout dialogue sessions. For each persona, we then create narrative trajectories specifying emotional progressions across conversational turns, simulating realistic emotional transitions while maintaining role consistency. Based on these personas and dialogue arcs, we generate specific scenarios with situational descriptions and emotional states that naturally evolve from previous interactions. Subsequently, we establish expected response characteristics for each scenario, including tone, content requirements, and emotional expression patterns. The final benchmark comprises 100 dialogue sessions with 8 scenarios each, totaling 800 evaluation instances that undergo manual verification to ensure quality. A compact summary of counts, metrics, models, and aggregation appears in Appendix A.20. All prompt templates used are detailed in Appendix A.14. Quality control measures, inter-annotator agreement analysis, and data leakage validation are detailed in Appendix A.21.